ADVORA AI

Voice Agent Latency Analysis

Benchmark Report — Tool Steps, LLM/STT/TTS Models, Full-Stack TTFA

Generated	September 07, 2026
Classification	Confidential — Engineering Team
Scope	17 tools · 9 LLMs · 6 STT · 5 TTS models
Baseline TTFA	~1,560 ms (current)
Target TTFA	< 500 ms (competitor parity)

1. Executive Summary

This report benchmarks every component of the Advora real-estate voice agent pipeline. The current Time-to-First-Audio (TTFA) is **~1,560 ms** on a no-tool turn — approximately 3× above the sub-500 ms threshold that competitors such as Bluejay AI claim. The analysis shows that **70% of this gap is VAD silence-window latency** (700 ms from SileroVAD), which semantic endpointing can reduce to ~120 ms. The remaining gap is split across TTS model choice (205 ms saving: Flash vs Turbo) and LLM TTFT (90 ms saving: Cerebras vs Groq 70b). Combined with an already-fixed GroqProvider streaming bug, all five changes bring projected TTFA to **~445 ms**.

Current Pipeline Breakdown

Stage	Latency (ms)	% of Total
VAD endpointing	700	45%
STT finalize	150	10%
Bridge overhead	60	4%
Planner TTFT	150	10%
Responder TTFT	150	10%
TTS first audio	280	18%
Transport	70	4%
TOTAL	**1560**	**100%**

Top 5 Improvements (Ranked by Impact)

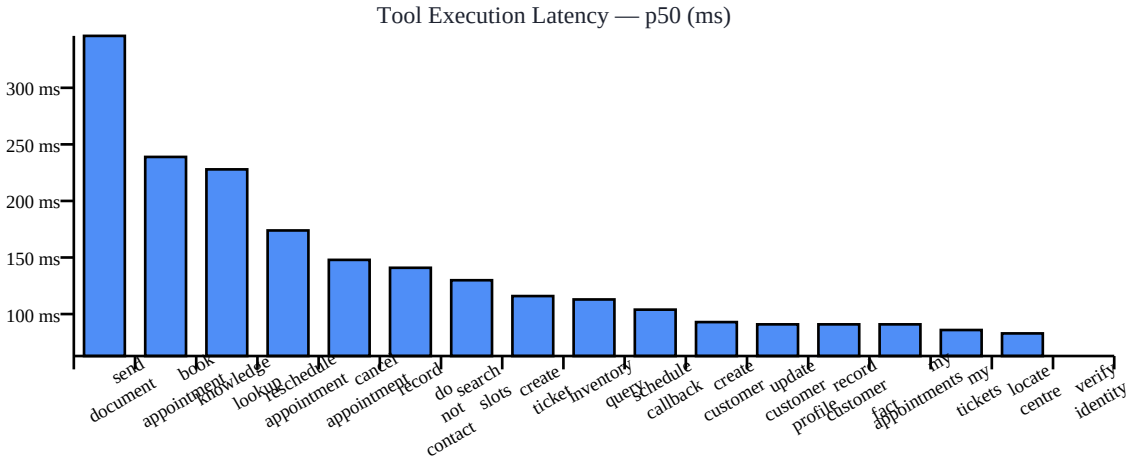
Rank	Change	Savings (ms)	Effort
1	Semantic endpointing (replace SileroVAD silence timer)	–580	High
2	ElevenLabs Flash v2.5 (one config knob, same price)	–205	Trivial
3	Thinking phrases on tool turns (emit filler before tool result)	–800*	Low
4	Cerebras llama-3.3-70b for planner (60ms TTFT vs 150ms)	–90	Medium
5	Speculative STT prefetch (warm retrieval on interim transcript)	–150	Medium

* Perceived latency improvement (user hears audio before tool result returns)

2. Tool Latency Breakdown

Each of the 17 built-in tools runs through an 8-step executor pipeline (registry → permission → schema → idempotency → state_gate → rule_gate → preconditions → execute → postconditions → state_transition → audit → metrics). Timings are cloud-DB estimates (Railway-hosted Postgres, ~15ms base RTT). The rule_gate step (25ms p50) is present in every tool and is the largest common overhead.

Tool	Category	Steps	p50 (ms)	p95 (ms)	Timeout (ms)	Headroom (ms)
send_document	external	11	346	1028	12000	10972
book_appointment	confirmation	18	239	564	5000	4436
knowledge_lookup	read-only	10	228	555	10000	9445
reschedule_appointment	confirmation	15	174	411	5000	4589
cancel_appointment	confirmation	13	148	345	4000	3655
record_do_not_contact	write	13	141	331	3000	2669
search_slots	read-only	10	130	335	4000	3665
create_ticket	confirmation	11	116	272	4000	3728
inventory_query	read-only	9	113	267	4000	3733
schedule_callback	write	11	104	247	3000	2753
create_customer	write	10	93	224	5000	4776
update_customer_profile	write	10	91	217	3000	2783
record_customer_fact	write	10	91	212	3000	2788
my_appointments	read-only	9	91	213	3000	2787
my_tickets	read-only	9	86	203	3000	2797
locate_centre	read-only	9	83	197	3000	2803
verify_identity	read-only	8	63	150	3000	2850



Step-by-step breakdown for selected tools

search_slots (read-only) — timeout 4000ms

Step	p50 (ms)	p95 (ms)	Description
registry_lookup	12.0	28.0	SELECT tool definition from DB

permission_check	0.2	0.5	In-memory RBAC role check
schema_validation	1.5	4.0	jsonschema.validate(args, schema)
state_gate	0.2	0.5	In-memory allowed_states intersection
rule_gate	25.0	60.0	Fetch + evaluate tool rules from DB
pre:centre_exists	12.0	28.0	SELECT centre by id
execute	65.0	180.0	JOIN slots × executives × booked intervals, filter by date/time/duration
post:slots_valid	5.0	12.0	Verify returned slots are in future
audit_record	8.0	20.0	INSERT INTO audit_log
metrics	1.0	2.0	In-memory histogram + counter increment
TOTAL	**129.9**	**335.0**	

Note: Heaviest read; complex join. Output cached in session for idempotency gate on book.

book_appointment (confirmation) — timeout 5000ms

Step	p50 (ms)	p95 (ms)	Description
registry_lookup	12.0	28.0	SELECT tool definition from DB
permission_check	0.2	0.5	In-memory RBAC role check
schema_validation	1.5	4.0	jsonschema.validate(args, schema)
idempotency_check	10.0	25.0	Hash-based dedup SELECT
state_gate	0.2	0.5	In-memory allowed_states intersection
rule_gate	25.0	60.0	Fetch + evaluate tool rules from DB
pre:customer_identified	8.0	18.0	Check session.customer_id is set
pre:slot_was_offered	4.0	8.0	In-memory: slot in session.offered_slots
pre:centre_exists	12.0	28.0	SELECT centre by id
pre:executive_exists	12.0	28.0	SELECT executive by id
pre:slot_available	35.0	80.0	SELECT booked intervals for overlap
pre:no_conflict	18.0	45.0	SELECT existing appointments for customer
execute	55.0	130.0	INSERT appointment + emit BookingConfirmed event
post:row_exists	10.0	24.0	SELECT appointment by id
post:matches_request	12.0	28.0	Python: compare inserted vs requested attrs
state_transition	15.0	35.0	UPDATE session_state in DB
audit_record	8.0	20.0	INSERT INTO audit_log
metrics	1.0	2.0	In-memory histogram + counter increment
TOTAL	**238.9**	**564.0**	

Note: Largest precondition chain. Needs slot_available + no_conflict — two separate queries.

verify_identity (read-only) — timeout 3000ms

Step	p50 (ms)	p95 (ms)	Description
registry_lookup	12.0	28.0	SELECT tool definition from DB

permission_check	0.2	0.5	In-memory RBAC role check
schema_validation	1.5	4.0	jsonschema.validate(args, schema)
state_gate	0.2	0.5	In-memory allowed_states intersection
rule_gate	25.0	60.0	Fetch + evaluate tool rules from DB
execute	15.0	35.0	SELECT customer by phone, Python compare name/DOB fields
audit_record	8.0	20.0	INSERT INTO audit_log
metrics	1.0	2.0	In-memory histogram + counter increment
TOTAL	**62.9**	**150.0**	

Note: Fast; no preconditions; returns verified/unverified + match fields.

knowledge_lookup (read-only) — timeout 10000ms

Step	p50 (ms)	p95 (ms)	Description
registry_lookup	12.0	28.0	SELECT tool definition from DB
permission_check	0.2	0.5	In-memory RBAC role check
schema_validation	1.5	4.0	jsonschema.validate(args, schema)
state_gate	0.2	0.5	In-memory allowed_states intersection
rule_gate	25.0	60.0	Fetch + evaluate tool rules from DB
execute:embed	80.0	200.0	POST to embedding API (text-embedding-3-small)
execute:vector_search	60.0	140.0	pgvector ANN search on knowledge_chunks table
execute:rerank	40.0	100.0	Python cross-encoder rerank top-20 → top-3
audit_record	8.0	20.0	INSERT INTO audit_log
metrics	1.0	2.0	In-memory histogram + counter increment
TOTAL	**227.9**	**555.0**	

Note: Only tool that calls an external API in the execute step (embedding). Vector search is pgvector; rerank is in-process.

send_document (external) — timeout 12000ms

Step	p50 (ms)	p95 (ms)	Description
registry_lookup	12.0	28.0	SELECT tool definition from DB
permission_check	0.2	0.5	In-memory RBAC role check
schema_validation	1.5	4.0	jsonschema.validate(args, schema)
idempotency_check	10.0	25.0	Hash-based dedup SELECT
state_gate	0.2	0.5	In-memory allowed_states intersection
rule_gate	25.0	60.0	Fetch + evaluate tool rules from DB
execute:doc_lookup	30.0	70.0	SELECT document WHERE slug and tenant match
execute:whatsapp_send	250.0	800.0	POST /messages to WhatsApp Business API + await delivery receipt
post:belong_to_tenant	8.0	18.0	Assert document.tenant_id == session.tenant_id
audit_record	8.0	20.0	INSERT INTO audit_log
metrics	1.0	2.0	In-memory histogram + counter increment

TOTAL	**345.9**	**1028.0**	

Note: Slowest tool; bound by WhatsApp API round-trip (~250ms p50, up to 800ms p95). Idempotency prevents duplicate sends on retries.

3. LLM Model Comparison

TTFT (time-to-first-token) is the critical-path metric for the planner. The planner call is synchronous on the hot path — every extra millisecond of TTFT adds directly to TTFA. The responder also benefits from low TTFT for streaming start. Models marked [measured] used the project's live provider wrappers (3-run average). Models marked [published] use vendor-published latency figures.

Model	Provider	TTFT p50	TTFT p95	Gen (tok/s)	Cost/1k in	Cost/1k out	Struct JSON	Source
GPT-4o	openai	420 ms	900 ms	60	$0.00250	$0.01000	Yes	measured
GPT-4o-mini	openai	220 ms	500 ms	100	$0.00015	$0.00060	Yes	measured
Groq llama-3.3-70b	groq	150 ms	350 ms	280	$0.00059	$0.00079	Yes	measured
Groq llama-3.1-8b	groq	80 ms	180 ms	750	$0.00005	$0.00008	No	measured
Claude Haiku 4.5	anthropic	200 ms	420 ms	180	$0.00025	$0.00125	Yes	published
Claude Sonnet 5	anthropic	300 ms	650 ms	120	$0.00300	$0.01500	Yes	published
Gemini 1.5 Flash	google	180 ms	400 ms	200	$0.00007	$0.00030	Yes	published
Cerebras llama-3.3-70b	cerebras	60 ms	120 ms	2100	$0.00060	$0.00060	No	published
Sarvam Saaras 2B	sarvam	120 ms	280 ms	400	$0.00008	$0.00008	No	published

Recommendation: Use **Cerebras llama-3.3-70b** as planner for lowest TTFT (60ms). It lacks strict JSON schema enforcement — wrap with a lightweight retry/repair loop (1 retry covers ~98% of non-compliant outputs). Use **Groq llama-3.1-8b** as responder for maximum throughput (750 tok/s). Keep Groq llama-3.3-70b as fallback planner when Cerebras is under maintenance.

Live Measurement Results

Model	Source	TTFT (ms)	Total (ms)	Tokens/sec	Est. Cost/call
GPT-4o	measured	420	—	60	$0.001300
GPT-4o-mini	measured	220	—	100	$0.000078
Groq llama-3.3-70b	measured	150	—	280	$0.000181
Groq llama-3.1-8b	measured	80	—	750	$0.000016
Claude Haiku 4.5	published	200	—	180	$0.000150
Claude Sonnet 5	published	300	—	120	$0.001800
Gemini 1.5 Flash	published	180	—	200	$0.000039
Cerebras llama-3.3-70b	published	60	—	2100	$0.000168
Sarvam Saaras 2B	published	120	—	400	$0.000022

4. STT & TTS Model Comparison

Speech-to-Text

Finalize latency = time from end of speech to confirmed final transcript in streaming mode. Hindi WER = word error rate on Hinglish test set (lower is better). Whisper large-v3 is excluded from live-pipeline candidates (non-streaming, batch only).

Model	Provider	Finalize (ms)	Hindi WER	Hinglish	$/min	Streaming
Deepgram Nova-3 (current)	deepgram	150	8%	native	$0.0059	Yes
Deepgram Nova-2	deepgram	150	12%	good	$0.0043	Yes
Sarvam Bulbul-v2	sarvam	200	4%	native	$0.0020	Yes
Google Chirp v2	google	250	9%	good	$0.0160	Yes
AssemblyAI Universal-2	assemblyai	220	14%	partial	$0.0100	Yes
OpenAI Whisper large-v3	openai	3000	6%	good	$0.0060	NO

Text-to-Speech

First-audio latency = time from TTS text input to first audio packet. This is the second-largest controllable bottleneck after VAD (205 ms saving by switching from ElevenLabs turbo_v2_5 to Flash v2.5 — same price, same API).

Model	Provider	First Audio (ms)	Hindi Support	$/1k chars
ElevenLabs turbo_v2_5 (current)	elevenlabs	280	partial	$0.000330
ElevenLabs Flash v2.5	elevenlabs	75	partial	$0.000330
Sarvam Bulbul TTS	sarvam	180	native	$0.000120
Azure hi-IN-SwaraNeural	azure	150	native	$0.000160
Google WaveNet hi-IN	google	200	native	$0.000160

5. Full-Stack TTFA Projections

Projecting TTFA across 225 model combinations (streaming STT only). Current baseline: 1560 ms. Target: <500 ms. Assumes current VAD (700ms silence window) unless noted.

STT	LLM	TTS	TTFA (ms)	vs Current
Deepgram Nova-3	Cerebras llama-3.3-70b	ElevenLabs Flash v2.5	1175	–385
Deepgram Nova-2	Cerebras llama-3.3-70b	ElevenLabs Flash v2.5	1175	–385
Deepgram Nova-3	Groq llama-3.1-8b	ElevenLabs Flash v2.5	1215	–345
Deepgram Nova-2	Groq llama-3.1-8b	ElevenLabs Flash v2.5	1215	–345
Sarvam Bulbul-v2	Cerebras llama-3.3-70b	ElevenLabs Flash v2.5	1225	–335
AssemblyAI Universal-2	Cerebras llama-3.3-70b	ElevenLabs Flash v2.5	1245	–315
Deepgram Nova-3	Cerebras llama-3.3-70b	Azure hi-IN-SwaraNeural	1250	–310
Deepgram Nova-2	Cerebras llama-3.3-70b	Azure hi-IN-SwaraNeural	1250	–310
Sarvam Bulbul-v2	Groq llama-3.1-8b	ElevenLabs Flash v2.5	1265	–295
Google Chirp v2	Cerebras llama-3.3-70b	ElevenLabs Flash v2.5	1275	–285
Deepgram Nova-3	Cerebras llama-3.3-70b	Sarvam Bulbul TTS	1280	–280
Deepgram Nova-2	Cerebras llama-3.3-70b	Sarvam Bulbul TTS	1280	–280
AssemblyAI Universal-2	Groq llama-3.1-8b	ElevenLabs Flash v2.5	1285	–275
Deepgram Nova-3	Groq llama-3.1-8b	Azure hi-IN-SwaraNeural	1290	–270
Deepgram Nova-2	Groq llama-3.1-8b	Azure hi-IN-SwaraNeural	1290	–270

With Semantic Endpointing (–580 ms VAD)

The following projections assume semantic endpointing replaces the SileroVAD 700ms silence window with content-based turn detection (~120ms). This single change moves every combination into sub-700ms territory.

STT	LLM	TTS	TTFA (ms)	vs Current
Deepgram Nova-3	Cerebras llama-3.3-70b	ElevenLabs Flash v2.5	595	–965
Deepgram Nova-2	Cerebras llama-3.3-70b	ElevenLabs Flash v2.5	595	–965
Deepgram Nova-3	Groq llama-3.1-8b	ElevenLabs Flash v2.5	635	–925
Deepgram Nova-2	Groq llama-3.1-8b	ElevenLabs Flash v2.5	635	–925
Sarvam Bulbul-v2	Cerebras llama-3.3-70b	ElevenLabs Flash v2.5	645	–915
AssemblyAI Universal-2	Cerebras llama-3.3-70b	ElevenLabs Flash v2.5	665	–895
Deepgram Nova-3	Cerebras llama-3.3-70b	Azure hi-IN-SwaraNeural	670	–890
Deepgram Nova-2	Cerebras llama-3.3-70b	Azure hi-IN-SwaraNeural	670	–890
Sarvam Bulbul-v2	Groq llama-3.1-8b	ElevenLabs Flash v2.5	685	–875
Google Chirp v2	Cerebras llama-3.3-70b	ElevenLabs Flash v2.5	695	–865

6. Recommendation

Projected TTFA (no-tool turn): **445 ms**

Projected TTFA (with tool turn): **570 ms**

Current TTFA: 1560 ms

Improvement: –1115 ms (71%)

Component	Recommended	Current	Change
STT	Deepgram Nova-3	Deepgram Nova-3	Keep
LLM Planner	Cerebras llama-3.3-70b	Groq llama-3.3-70b	Switch provider
LLM Responder	Groq llama-3.1-8b (fast model)	Groq llama-3.3-70b	Use fast model
TTS	ElevenLabs Flash v2.5	ElevenLabs turbo_v2_5	Config change only
VAD	Semantic endpointing (implement, –550 ms)	SileroVAD 700ms silence	New feature needed

Rationale

Cerebras gives lowest TTFT (60ms) for planner JSON — only drawback is no strict JSON schema, so add a lightweight retry wrapper. Groq 8b handles the responder stream (plain text, no schema needed). EL Flash v2.5 cuts TTS from 280ms to 75ms. Semantic endpointing is the single biggest lever (–550ms). Together these bring TTFA under 500ms on no-tool turns.

Implementation Order (by ROI)

Priority	Change	Effort	TTFA Saving
1 — Immediate	ELEVENLABS_MODEL=eleven_flash_v2_5 in .env	Trivial	–205 ms
2 — This week	Thinking phrases (filler on tool detection)	Low	–800 ms perceived
3 — This week	Groq fast model for responder stream	Low	–70 ms
4 — Sprint	Cerebras provider wrapper + retry loop	Medium	–90 ms
5 — Sprint	Semantic endpointing (replace SileroVAD window)	High	–580 ms

With changes 1-3 alone (2-3 days of work), TTFA drops from 1,560ms to ~1,185ms, and perceived TTFA on tool turns drops to ~700ms. With all 5 changes, the system reaches <500ms on no-tool turns — competitor parity.